

# HealthConnect Clinic, Week 8

HC-POD Final Integration Record (Data Analytics Track, DA-only)

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Documented as a DA-only walkthrough, since this HC-POD is worked solo. The DS track collaborator is a separate intern's track referenced by their artefacts only, not named.

## Integration Record: Data Analytics → DS Track Collaborator

|                                               |                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Track collaborated with</b>                | The DS track collaborator (a separate intern's track; their artefacts were used, not named).                                                                                                                                                                                                                                     |
| <b>Dependency</b>                             | The DS track collaborator's <code>double_risk_flag</code> feature in <code>train_improved.py</code> claims to encode the DA-validated double-risk finding (long lead time plus 2+ prior no-shows, roughly 73% risk in roughly 4.7% of bookings).                                                                                 |
| <b>Output received</b>                        | The feature code and its comment, which cited the DA finding as rationale but coded the threshold at lead time $\geq 46$ days.                                                                                                                                                                                                   |
| <b>Output provided</b>                        | The DA-validated segment definition and figures (lead $\geq 31$ days and prior no-show bracket $\geq 2$ , 4.66% of bookings, 73.0% no-show rate), independently reproduced from the shared prepared dataset, plus a handoff note and evidence package.                                                                           |
| <b>Final integration activity</b>             | Recomputed the segment under both thresholds directly from the shared prepared dataset, then retested model performance under both a Random Forest and the exact Week 6 candidate Logistic Regression. The corrected threshold is now confirmed ready to merge into <code>train_improved.py</code> for Week 8 final integration. |
| <b>What changed</b>                           | The double-risk segment threshold was corrected from $\geq 46$ to $\geq 31$ days. The corrected definition is documented as the standard for the final model and this report.                                                                                                                                                    |
| <b>How the change improved the solution</b>   | Removed a real inconsistency between the DS model's documentation and its code, confirmed under two model types that the threshold choice does not affect model performance, and gave the DS track collaborator a corrected, evidence-backed feature definition for final integration.                                           |
| <b>Evidence</b>                               | <code>reports/cross_track_evidence/</code> (segment retest CSVs and output logs under both model types), Week 7 Analytics Testing & Refinement Report Section 6, and the <code>handoff/ds_track_collaborator/</code> package.                                                                                                    |
| <b>Contribution to the final presentation</b> | The double-risk segment (4.66% of bookings, 73.0% no-show rate) is presented as a headline finding and the basis for the proposed call pilot. The correction itself is used as an example of testing catching a real cross-track discrepancy.                                                                                    |

Note: this HC-POD is being completed solo. The record above is the one real, artefact-level exchange this track had during Week 7 to Week 8: a tested correction to the DS track collaborator's code. No other track dependency existed for this component this cycle.